Chih-Hsuan (Bella) Yang

Research on Reliable and Efficient AI Agents

bellayang@anl.gov | chihhsuan-yang.github.io
GitHub | Hugging Face | Google Scholar | AgentsSci | LinkedIn

Postdoctoral researcher at Argonne National Laboratory studying large language model (LLM) agents and multi-agent systems. Develops evaluation harnesses and controlled experiments to understand when collaboration, feedback, and memory improve reasoning, including agentic AI for Science.

Research Contributions and Accepted Publications

Precise but Uncoupled: Reviewer Precision Does Not Guarantee Critique Uptake in Multi-Agent Math Reasoning

First author. NeurIPS Main Track, accepted. 2026

Designed controlled analyses separating reviewer error detection, a solver's response to critique, and verified answer repair. Identified why accurate review alone need not improve final outcomes, informing agent feedback design.

Paper and project | Code | Data

LLMs Can Predict Failure Risk, But Struggle to Predict Which Collaboration Protocol Pays Off: Cost-Aware Protocol Routing Across Reasoning Tasks

First author. Findings of EMNLP, published. 2026

Compared four reasoning protocols on a 4,181-problem primary math benchmark, with robustness checks across mathematical and scientific reasoning settings and two solver families. Distinguished failure prediction from protocol-specific value; released matched outcomes, evaluation code, and a trained router.

Paper | Code | Data and router

BioTrove: A Large Curated Image Dataset Enabling AI for Biodiversity

First author. NeurIPS Datasets and Benchmarks, Spotlight. 2024

Co-led a scientific data effort yielding 161.9 million research-grade images across approximately 366,600 species, with vision-language models and zero-shot biodiversity benchmarks.

MolGen-Transformer: A Molecule Language Model for the Generation and Latent Space Exploration of Organic Molecules

First author. Computational Materials Science 269, 114549. 2026

Developed transformer-based molecular generation and controlled latent-space exploration for molecular and materials discovery.

Current Research Directions

Study how roles change feedback access and failure recovery, and measure task-conditioned memory utility using paired evaluations with and without a memory while holding other context fixed.

Research Experience and Mentoring

Postdoctoral Researcher, Argonne National Laboratory Jul 2025 - Present
Lead research on agent collaboration, evaluation harnesses, feedback, memory, and inference efficiency. Mentor four students and coordinate research questions, controlled experiments, manuscripts, and reproducible code and data releases.

Visiting Scientist, Corteva Agriscience Sep 2024 - Jan 2025
Trained preference models from scratch using real experimental outcomes to ground rewards. Adapted multiple language models with Direct Preference Optimization (DPO) and Proximal Policy Optimization (PPO); incorporated scientist preferences when model judgments were uncertain.

AI Resident, X, Google LLC Sep - Dec 2023
Developed multi-agent LLM frameworks representing stakeholders with different interests. Investigated information flow, coordination, communication, and bidding mechanisms.

AI Graduate Researcher, Iowa State University Jan 2020 - Jun 2025
Researched molecular language modeling, multi-fidelity materials design, neural solvers for partial differential equations, and multimodal biodiversity models; co-led the BioTrove data effort.

Research Methods and Implementation

Matched protocol comparisons, trace analysis, benchmark design, preference and reward modeling, reinforcement learning from human feedback (RLHF), DPO, and PPO. Python, PyTorch, Hugging Face, NumPy, pandas, scikit-learn, pytest, Git, and high-performance computing experimentation.

Selected Manuscripts Under Review

Wrong but Useful: Trajectory Value Beyond Answer Correctness in Multi-Agent Messages
First author. Submitted to AAAI 2027, under review; public preprint. 2026

Beyond Division of Labor: How Roles Shape Feedback and Recovery in Multi-Agent Systems
First and corresponding author. AAMAS 2027, under review. 2026

What Is This Memory Useful For? Measuring Task-Conditioned Utility in Long-Term Agent Memory
First and corresponding author. AAMAS 2027, under review. 2026

Education

Ph.D., Mechanical Engineering, Iowa State University 2025
Co-major in Computer Engineering; minor in Applied Mathematics.

M.S., Computer Science, Iowa State University 2025

B.S., Bioenvironmental Systems Engineering, National Taiwan University 2019